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Preview

Advancing the Integration of Biosciences Data Sharing to Further Enable Space Exploration

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Understanding the impact of space exploration remains biologically elusive. Cell Press is dedicating this month to spaceflight (Afshinnekoo et al., 2020), with the open science NASA GeneLab database enabling the study revealing mitochondria as a key biological feature from spaceflight (da Silveira et al., 2020).

Increasingly, many life sciences research questions necessitate that data be reused, for both exploration of new hypotheses and validation of existing ones. The ability to mine multiple datasets at once, evaluating how experimental factors affect the endpoints of standard assays, has become a new phase in the pursuit of gaining knowledge and putting forward new hypotheses. By integrating data and biospecimen resources from past, present, and future spaceflight and ground-based analog experiments, we can now utilize data tools such as artificial intelligence (AI) and machine learning (ML) methodologies to infer new biology. This effort can enable interdisciplinary life science spaceflight research across spaceflight missions and across spaceflight hazards (Goswami et al., 2013), along with precision aerospace medicine and further systems biology analyses. This approach requires an adoption of the FAIR principles (Wilkinson et al., 2016), along with versatile application programming interfaces (APIs). The result is a powerful set of resources, which enables data search queries, visualization, analysis, and, eventually, modeling.

NASA non-omics data and non-human biospecimens are currently available through the Life Sciences Data Archive (LSDA), a searchable NASA-federated data archive. Ames Life Sciences Data Archive (ALSDA) is the Ames Research Center node of the LSDA and archives biospecimens and data from mammalian and microbial experiments (metadata, experiment raw assay data, hardware specifications, telemetry, scientific reports, and imagery). Biospecimens are housed within the Institutional Scientific Collection at Ames Research Center (ISC-ARC), a biorepository facility aligned with best practices and standards (Campbell et al., 2018). Currently, more than 32,000 mammalian biospecimens are available for request by researchers, with samples mostly from Shuttle, International Space Station (ISS), and ground-based flight analog investigations.

A panel of experts from the National Academy provided guidance in 2011 to the NASA Biological and Physical Sciences (BPS) Division (formerly known as the Space Life and Physical Sciences Research and Applications Division), suggesting the need for greater informatics technologies to meet the growing needs of researchers. They noted a strong desire by investigators for systems biology approaches and suggested creation of an extensive and well-organized database, focused on quantitative biological data such as genomics. In response, BPS funded the development of the GeneLab Project, which, in less than 5 years, has become the largest online platform to mine and visualize omics, molecular, and related data from spaceflight-relevant experiments (Berrios et al., 2020).

Integration efforts need to reinforce existing data linkages between ALSDA and GeneLab. Currently, GeneLab experimental factors (metadata) are crosslinked with LSDA. Data provided by ALSDA to GeneLab include environmental data and radiation dosimetry from spaceflight experiments. GeneLab investigators can then evaluate how these factors affect molecular pathways. ISC-ARC offers the iterative ability to analyze archived biospecimens with novel assays to test emerging hypotheses from omics analysis. GeneLab has been awarded many ISC-ARC biospecimens to generate omics data, which are open access through GeneLab's online portal. Approaches are underway to further harmonize metadata between GeneLab and ALSDA to expand the breadth of useable data for scientists, linking omics datasets with higher-order physiological datasets from the same samples and experiments. Due to the volume and complexity of spaceflight data, it is important to adopt good data management guidelines and enhance programmatic access to support discovery.

Science users of GeneLab benefited from the project's early compliance with the FAIR principles (Wilkinson et al., 2016): i.e., data must be findable, accessible, interoperable, and reusable. As a result of the early adoption of FAIR, GeneLab has surpassed mainstream databases in its "FAIRness" metrics (Berrios

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et al., 2018). GeneLab data are easily "findable" with each dataset being assigned persistent IDs such as DOIs (digital object identifiers), rich metadata with the adoption of the ISA-Tab standard (https://isa-tools.org/), and careful curation of metadata into JSON-LD/Schema.org (enabling efficient keyword searches). A strong API has been essential to providing "accessibility" to GeneLab data, while DOI usage also ensures that metadata remain permanently accessible. The GeneLab data submission portal was designed to control vocabulary used for experiment descriptions. This provides metadata consistency, required for "interoperable" data, along with the API facilitating communication with other database-ontology resources. "Reusability" is achieved by publishing data with clear provenance lineage for metadata, alongside community-derived standards.

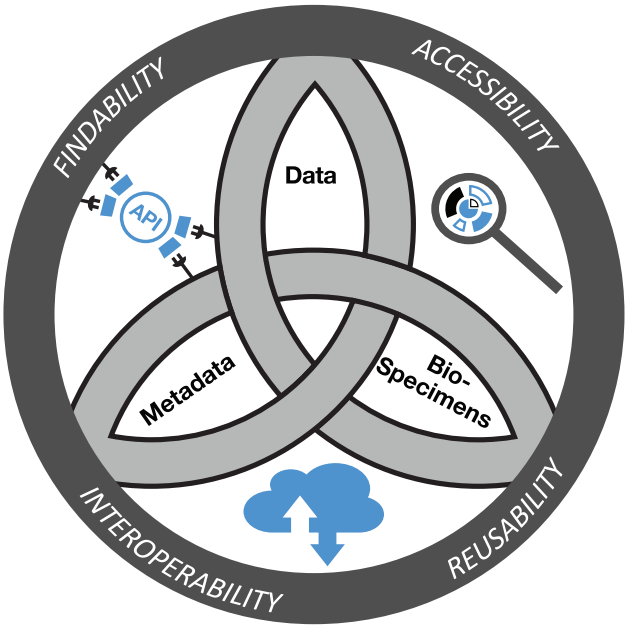


Figure 1. Integration of Data, Metadata, and Biospecimens

The GeneLab API bridges metadata with underlying experimental data, with the ISA-Tab format enforcing a strict hierarchical model for metadata storage. Relying on harmonized ISA-Tab metadata such as factor values and characteristics, the API enables database interrogation down to the sample level, through logical queries for metadata fields and values, and retrieval of experimental data associated with these samples. Results are returned in a table format, ready for ingestion by programming languages and ready for data science discovery environments. Cross-dataset analyses and data visualization (e.g., differential expression analyses, sample-gene clustering, pathway enrichment analyses) are enabled, providing a toolset for investigating the impact of the environment of spaceflight in an integrated multi-dataset, multi-omics paradigm. Since the API essentially maps metadata onto all types of files in a dataset, it can also handle ALSDA categories.

Integration of data from GeneLab and ALSDA will enable spaceflight health risk modeling. Current health risk models are derived from *in vitro* and *in vivo* experiments conducted either in space or via ground analogs. However, it is difficult to combine data to inform biological health risks due to data heterogeneity and lack of harmonization of metadata. Such an endeavor is highlighted in Cell Press publications this month, dedicating several articles to meta-analysis, revealing, for example, the importance of mitochondria to the space environment biological response (da Silveira et al., 2020). A comprehensive review covering our current knowledge on the fundamental biological features of spaceflight illustrates nicely such analytical challenges (Afshinnekoo et al., 2020). Next-generation data science methodologies such as AI and ML will enable us to further leverage such heterogeneous datasets, as recently illustrated on omics data taken on the ISS (McIntyre et al., 2019). By classifying experimental factors into digital objects and by transforming all raw data into quantitative endpoints with a common workflow, one can mathematically relate space factors (e.g., microgravity, radiation, carbon dioxide levels) with biological endpoints (Beheshti et al., 2019). This baseline of predicted norms for biological changes, again using AI and ML, can be incorporated into spaceflight health monitoring and diagnostic device development, automating and personalizing aspects of spaceflight health (Roda et al., 2018). The integration of data, and metadata, and biospecimens is essential to utilizing the 40+ years of archived NASA Life Sciences data (Figure 1), and this can be done by extending GeneLab's data system to combine omics with non-omics ALSDA data (cellular, tissue, organ, and whole-organism levels). All data would then benefit from applied FAIR principles, and users would enjoy a single point of entry for all space biology data submission. Such an approach would expand metadata harmonization beyond omics into higher-order datasets, also aiding the development of next-generation spaceflight health risk models.

This open access science perspective invites investigators to participate in a transformative collaborative effort for interpreting spaceflight effects by integrating omics and physiological data to the systems level. This pursuit will accelerate knowledge acquisition, interdisciplinary research, predictive modeling, and, eventually, precision space medicine.

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